Can I equalize AGM batteries?

This question comes up every day. Can I equalize AGM batteries? The short answer is Yes! Let's start at the beginning on what equalizing is, what it actually does and how it can be beneficial to extending your battery life. Please keep in mind that not all AGM batteries are built equally. Our batteries are hand built and have large inter-cell connections, allowing them to be equalized. If you do not have a Lifeline AGM deep cycle battery, please consult the manufacturers recommendations before you start this process.

Batteries are pretty simple. You put energy into them, they store it, and then they give you energy when you need it. The most important part of that cycle is making sure that you put back all the amperage you have taken out. As a battery manufacture we understand that this is not always feasible. Certain applications make it almost impossible to recharge back to 100% every time. Equalizing can help.

Lead Acid Batteries Sulfate

Let us start by telling you all lead acid batteries sulfate. This is a natural occurrence every time you discharge. Keep in mind that Wet Cell, GEL and AGM batteries are all lead acid batteries. They have different ways on containing acid but they are all a lead acid based structure. When you have a fully charged battery it is lead and sulfuric acid (Active material). When you start discharging a battery, the chemical reaction between lead and sulfuric acid provides energy for your application. The byproduct of this reaction is lead sulfate. Lead sulfate is a soft material that naturally covers the plates, both positive and negative, when the battery is being discharged. The deeper the discharge, the more lead sulfate will cover the plates. Once the load is turned off and it is time to recharge, lead sulfate is easily converted back into lead and sulfuric acid. This is why it is important to charge batteries back to 100%. If you don't you will not convert all the lead sulfate back into the active material. If a lead acid battery is not immediately recharged, the lead sulfate will begin to form hard crystals, which cannot be reconverted by standard charging voltages. The longer this goes on, the harder it is to get all of your capacity back. We can sum this up with an example.

100 AMP Hour Battery Example

Let's say you have a 100 amp hour battery. You discharge 50 amps out of it. You only recharge back to 85 amps (85% recharged). The result is the last 15% of the plates never got converted back into active material. That 15% remains lead sulfate. Now you no longer have a 100 amp battery, you only have an 85 amp battery. As this continues the lead sulfate will get harder and harder. It will also start covering more of the plates, soon resulting in an 84 amp battery, then 83,82,81,80,etc........ If this goes on too long the lead sulfate will become so hard, even equalizing will not bring the battery back to its original condition. Think of a battery like a fuel tank. If you get 100 miles out of 10 gallons of gas. Let's say the next time you fill up your gas tank you only put 8.5 gallons in it. You can't expect to get 100 miles out the tank that you only filled up to 85% of its capacity.

With all that said, let's talk about equalizing. Equalizing is a high voltage charge that will convert lead sulfate back into active material. This has to be done within a reasonable amount of time for it to work. As stated earlier, the longer it goes on the harder it becomes to remove. Our batteries should be equalized at 15.5-16.3 volts for 6-8 hours. Depending on the level of sulfation, you may need to perform this process 2-3 times. You must perform an equalizing charge after you have gone through a regular charge first and the batteries have reached float charge. After that you can start the equalizing. The amperage doesn't necessarily matter. The idea is to get the amperage that the batteries need back into them. Sometimes they take 15-20 amps and sometimes they only take 5 amps. Every battery and application are different in their own right. You can usually tell if your batteries are sulfated when the open circuit voltage (OCV) is low. For example, a fully charged battery should be 12.8 volts are higher. A sulfated battery will be "finished" charging, yet only have an OCV of 12.2-12.5 volts. This is a good indication that the batteries need to be equalized.

AGM Battery Charging

At Lifeline we understand that some applications, especially sailing vessels, cannot always fully recharge every day. As a result you will need to equalize. Your equalizing routine will depend on a lot of factors. We are here to help you custom design a custom equalizing routine based on your application and habits to make your batteries last as long as possible. We personally worked with John Harries, which owns and operated a sailing website www.morganscloud.com. After two rounds of AGM batteries failing within two years, John called us for advice. We provided John with two new batteries and a custom equalizing routine. His new batteries lasted him 5+ years in the same vessel.

In close, all of us here at Lifeline are willing to assess your application and help you with a successful battery charging routine that will make your batteries last as long as possible. Ideally you should recharge back to 100% every time you use your batteries. If you cannot do this equalizing can help.